

Why do AI data centers need clean water?

Posted by u/Dry-Poetry-8708 • 0 points • 73 comments

I know why they need **water**, I get that it's for cooling, what I don't get is why they need **clean** water.

I keep hearing people talk about how these data centers are taking up all the potable water for cooling, but like, why does it have to be the clean water? I don't see why grey or polluted, non-potable water would be any less cooling, it's not like these machines are drinking, and it's not like the cleanliness of the water affects its temperature abilities (or does it?)

I fully get I might be misunderstanding something? Maybe they already do use dirty water?

Is it just the run off polluting water and maybe they don't utilize the clean stuff at all? Is that **all** it is? A lot of the debates are indicating that they absorb the potable water, why?

Also, please note, I'm not posting this to debate AI ethics, there are plenty of other opportunities for that debate. I expect it will inevitably happen anyway, but that's not my intent here.

Maybe the answer I get could be **used as fuel** for a **future** ethics debate, but for now, I just want a plain, non-political straight factual answer to the question.

I've done many a Google with a serious lack of a straight answer, so here I am.

Comments

u/cantsay • 1 points • 2026-03-26 00:42 UTC

We need to legislate that these motherfuckers use treated sewage water or fracking runoff or something like that. Boohoo the fucking cost, do you want to build it or not.

u/StrangeReason • 1 points • 2026-02-22 16:27 UTC

Too bad we can't do liquid nitrogen.

u/Dry-Poetry-8708 • 1 points • 2026-02-22 16:32 UTC

Perhaps a silly question, but why not? Sorry if that's an ignorant thing to ask. I might be totally out of the loop here. Is that the stuff that shatters things? Is that why? Is this a joke about shattering

the data centers?

Please excuse my literal mindedness.

u/EmergencyAd7567 • 1 points • 2026-02-19 17:57 UTC

They DONT. It's just easier to tap into a water main. Long term, it would be cheaper to set up capture lines and keep cycling the same water through the system, like nuclear facilities do, and then they could just use non potable water. Problem is that even the people owning and operating these facilities don't see it as a permanent thing, so they don't want to "waste" the money on establishing infrastructure

u/Previous-Battle-9710 • 1 points • 2026-02-06 14:06 UTC

Why can't I use dirty coolant to keep my engine running cool 🤔

u/lreeb • 2 points • 2026-01-11 11:08 UTC

The actual cooling elements on the chips typically have really fine channels the water flows through, and these would quickly clog up with unclean water. Dirty water could also cause other stuff to build up in various locations of the cooling systems.

Of course, they could cycle the same, purified water over and over again, but then they would need to cool it down, and they don't want to pay for that when they can just keep using cool, clean drinking water.

u/corgis_are_awesome • 2 points • 2026-01-11 08:46 UTC

Most of the "water usage" by data centers is fear mongering and misinformation spread by anti-ai luddites.

Data centers use closed-loop water cooling and they recycle most of the water. Only a small fraction of the water is evaporating. It's not like they are just pouring cold water over the servers and down the drain.

Golf courses currently use/waste 3-5 times more water than all of the data centers in the entire USA combined.

The misinformation about this topic really annoys me.

u/Alternative_Novel_51 • 1 points • 2026-01-21 07:38 UTC

do you have a source on the golf course thing?

u/corgis_are_awesome • 1 points • 2026-01-27 10:33 UTC

Sure. Have a read:

<https://andymasley.substack.com/p/the-ai-water-issue-is-fake>

u/Alternative_Novel_51 • 1 points • 2026-01-28 07:59 UTC

Full disclosure, I did not read the full article, Andy Masley uses AI to "write" his articles, I'm not going to bother to read something he didn't even bother to write.

I did read the section on AI water usage as compared to golf water usage, in which he decides to exclude the water used to generate

electricity to power these sites. The mental gymnastics required to make this private definition are ridiculous. It's true that AI water usage is over-reported, but this guy is just as dishonest as the people he's criticizing.

u/SorenDarkSky • 1 points • 2026-01-11 00:35 UTC

you know, wouldn't all that heat in data centers be good for thermal distillation in a desalination plant? dang if only someone could figure out how to get them to work together

u/Bicyclebillpdx_ • 1 points • 2026-01-11 00:03 UTC

Most current data center designs use evaporative cooling. Fiberglass media that you saturate with water and blow air through. They use recycled water in some cases where it's available. It's important to keep the total dissolved solids (tds) low to avoid fouling of the media. Also important to keep clean to avoid bacteria from growing on it. Even though it's cooling servers, humans still need to be able to breath it and not get sick. Look up legionnaires disease for reference.